Aditya Ganguli

Mississauga, Ontario | 647-271-1929 | adi.p.ganguli@gmail.com

EDUCATION

McMaster University — Hamilton, Ontario
Bachelor of Computer Engineering — Expected May 2029
Cumulative GPA: 3.86/4.0; Dean's List 2024-2025, 2025-2026
Relevant Coursework: Data Structures & Algorithms, Embedded Systems Design, Digital Logic, Computer Architecture

WORK EXPERIENCE

Zurich Canada Holdings Limited — Toronto, Ontario
IT Core Systems IV Intern — May 2025 – Aug 2025

- Led the design and deployment of a Datadog error monitoring system across multiple PolicyCenter environments, tracking a comprehensive array of runtime errors to enhance observability and incident response.
- Engineered an Excel-VBA data pipeline to aggregate, sort, and analyze 14,000+ Guidewire policies, generating structured summary reports across multiple datasets.
- Engineered data-driven Jira dashboards for Commercial Auto and Middle Market teams to track sprint efficiency and bottlenecks, improving delivery and decision-making.

PROJECTS

Ashida Wii – A Custom Portable Nintendo Wii Console — Sept 2026

- Designed a fully functional handheld Wii console by precision-trimming an OEM Wii motherboard, relocating critical power traces, and integrating a dedicated Power Management System (PMS) with USB-C charging in a custom 3D-printed shell.
- Patched low-level system firmware and configured a custom bootloader on the Wii to bypass legacy optical drive dependencies, streaming disk images directly from onboard flash storage while optimizing background power consumption.
- Micro-soldered magnet and silicone wiring to route 480p VGA video, amplified stereo audio, and harvested OEM controller parts as inputs for custom breakout boards managing controls.

GBU - Gameboy Emulator written in C++ — May 2026

- Engineered a low-level Game Boy hardware emulator from scratch in C++, utilizing SDL2 to handle real-time graphics rendering, window management, and peripheral input mapping.
- Simulated the internal computer architecture of the Sharp LR35902 CPU, implementing cycle-accurate opcode decoding, register files, and an integrated Memory Management Unit (MMU).
- Developed memory bank switching (MBC) controllers and persistent cartridge save-state logic to accurately dump and load SRAM data for AAA titles like Pokémon and Super Mario Land.

Small-Scale LiDAR Scanner — Apr 2026

- Developed a portable LiDAR spatial scanner using a Texas Instruments MSP432 microcontroller, writing the core system control firmware in C within Keil uVision.
- Integrated I2C communication at 100 kbps to interface with a Time-of-Flight distance sensor and implemented a 115200 bps UART link to stream real-time data to a host PC.
- Programmed a stepper motor to execute precise 360-degree, 64-point indoor boundary scans, utilizing a MATLAB script to parse and convert polar coordinates into volumetric 3D wireframe models.

VOLUNTEERING

Mississauga Taekwondo — Mississauga, Ontario
Volunteer Instructor — Jul 2020 – Aug 2023

- Dedicated 3+ years to instructing students in martial arts techniques while mentoring youth to develop discipline, structural focus, and leadership skills.

Ontario Konkani Association — Mississauga, Ontario
Youth Coordinator Member — May 2022 – May 2024

- Coordinated youth activities for cultural and community events, including festivals, picnics, and charity initiatives.

ADDITIONAL

Technical Skills: Advanced in Quartus, ModelSim, LTSpice, PSpice, Autodesk Inventor/AutoCAD, Git, KiCad, Fusion 360, Keil uVision, Soldering, Arduino, Raspberry Pi, ARM Based Microcontrollers (MSP432E401Y, ESP32, STM32)
Programming Languages: Advanced in C/C++, Java, JavaScript, Python, Verilog, VHDL, MATLAB, HTML, CSS, Assembly